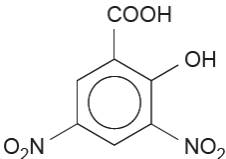


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)	Indicative content - flasks or beakers - suitable sizes - use of a fume cupboard - use of a thermometer and method of maintaining 60 °C temperature - stirring - filtration - filtered crystals washed with water - until filtrate is colourless (AO3) - reaction flask washed with water and contents added to filtering - apparatus - crystals removed and dried in a warm oven / at room temperature / below the melting temperature of the product (AO3)	4		2	6		6
			5-6 marks Specific detail from the question has been used; reference to apparatus sizes, use of a fume cupboard for the reaction and washing the filtered crystals <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i> 3-4 marks Mainly generic description; use of a fume cupboard <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Reference to reaction and filtration stages <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		mole ratio 1:1 moles of 2-hydroxybenzoic acid used = $\frac{4.00}{138.06} = 0.0290$ (1) theoretical moles of nitroacid = 0.290 theoretical mass of nitroacid = 5.306 / 5.31 g (1) percentage yield is 41 therefore mass obtained = $\frac{5.31 \times 41}{100} = 2.18$ g (1)		3		3	2	
		(iii)		relative mass of compound J without two X groups ($C_7H_6O_3$) = 136 relative mass of two X groups is 228 – 136 = 92 each X group has mass of 46 (1) award (1) for either of following - X must contain 2 oxygen atoms (as 7 in total) atoms therefore remainder is 14 - other atom must be nitrogen and X is NO_2 - structure of compound J 			2	2		
		(iv)		lower temperature / lower concentration HNO_3 / lower volume of aqueous HNO_3 / less heating time			1	1		
	(b)			2-hydroxybenzenecarboxylic acid will react with $NaHCO_3$ / Na_2CO_3 to give effervescence	1			1		1
				Question 11 total	5	3	5	13	2	7